

SECOND ASSIGNMENT :

Selection 1: Lists a. Create a list: Create a list containing the numbers 1 through 15. Print the list.

```
In [23]: list = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]
         print(list)
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15]
```

b. List of strings : Create a list of your five favorite fruits. Print the list.

```
In [24]: fruits = ["Mango", "Banana", "Litchi", "Apple", "Grapes"]
         print(fruits)
```

```
['Mango', 'Banana', 'Litchi', 'Apple', 'Grapes']
```

c. Accessing elements : Given the list [10, 20, 30, 40, 50], print the first and last element using positive and negative indexing.

```
In [25]: num = [10, 20, 30, 40, 50]
         print(num)
         print(num[0], num[-1])
```

```
[10, 20, 30, 40, 50]
10 50
```

d. List Length : Create a list of any 5 items and print its length using len() function:

```
In [26]: items = ["Apple", "Banana", "grapes", "Mango", "Papaya"]
```

```
In [27]: print(len(items))
```

```
5
```

e. Appending Elements: Start with an empty list and append the numbers 1, 2, and 3. Print the list.

```
In [28]: lst = []
         type(lst)
```

```
Out[28]: list
```

```
In [29]: lst.append([1, 2, 3])  
print(lst)
```

```
[[1, 2, 3]]
```

f. Inserting an Element: Given a list [1, 3, 4], insert the number 2 at the correct position so that the list becomes [1, 2, 3, 4].

```
In [30]: lst = [1, 3, 4]  
lst.insert(1, 2)  
print(lst)
```

```
[1, 2, 3, 4]
```

g. Removing an Element: Remove the number 3 from the list [1, 2, 3, 4, 5] using a list method and print the new list.

```
In [31]: lst1 = [1, 2, 3, 4, 5]  
lst1.remove(3)  
print(lst1)
```

```
[1, 2, 4, 5]
```

h. Popping an Element: Given the list [10, 20, 30, 40], pop the last element and print the element and the updated list.

```
In [32]: lst2 = [10, 20, 30, 40]  
lst2.pop()  
print(lst2)
```

```
[10, 20, 30]
```

i. Slicing a List: Given the list [0, 1, 2, 3, 4, 5], print a slice that contains the elements from index 2 to 4.

```
In [33]: lst3 = [0, 1, 2, 3, 4, 5]  
print(lst3[2 : 5]) # In slicing ending index is exclusive
```

```
[2, 3, 4]
```

j. List Concatenation: Concatenate two lists, e.g., [1, 2, 3] and [4, 5, 6], and print the resulting list.

```
In [34]: a = [1, 2, 3]  
b = [4, 5, 6]
```

```
concat = a + b  
print(concat)
```

[1, 2, 3, 4, 5, 6]

k. Repeating a List: Create a list [1, 2] and print the list repeated three times.

```
In [35]: c = [1, 2]  
print([c]* 3)
```

[[1, 2], [1, 2], [1, 2]]

l. Copying a List: Create a copy of a given list and print both the original and the copy.

```
In [36]: d = [1, 2, 3, 4]  
dup = d.copy()  
print(d)  
print(dup)
```

[1, 2, 3, 4]

[1, 2, 3, 4]

m. Clearing a List: Given any list, use a method to clear all its elements and then print the empty list.

```
In [37]: e = [1, 2, 3, 4]  
e.clear()  
print(e)
```

[]

SELECTION 2 : TUPLE

a. Create a Tuple: Create a tuple containing the numbers 1, 2, and 3. Print the tuple. Tuple of Strings:

```
In [38]: tup = (1, 2, 3)  
print(tup)
```

(1, 2, 3)

b. Tuple of Strings: Create a tuple of three different color names and print it.

```
In [39]: color = ("Red", "Orange", "Crimson")
print(color)
```

('Red', 'Orange', 'Crimson')

c. Accessing Tuple Elements: Given the tuple (10, 20, 30, 40), print the second element.

```
In [40]: num = (10, 20, 30, 40)
print(num[1])
```

20

d. Tuple Slicing: Using the tuple (0, 1, 2, 3, 4), print a slice that contains elements from index 1 to 3.

```
In [41]: num = (0, 1, 2, 3, 4)
print(num[1:4]) # In slicing last index is exclusive
```

(1, 2, 3)

e. Concatenating Tuples: Concatenate two tuples, e.g., (1, 2) and (3, 4), and print the result.

```
In [42]: a = (1, 2)
b = (3, 4)
concat = a + b
print(concat)
```

(1, 2, 3, 4)

f. Tuple Unpacking: Store the tuple ("Alice", 25, "New York") into three variables and print them.

```
In [43]: name, age, city = ("Alice", 25, "New York")
print(name)
print(age)
print(city)
```

Alice

25

New York

g. Convert List to Tuple: Convert the list [1, 2, 3, 4] into a tuple and print the tuple.

```
In [44]: lst = [1, 2, 3, 4]
         tup = tuple(lst)
         print(tup)
```

(1, 2, 3, 4)

h. Counting Occurrences: Given the tuple (1, 2, 2, 3, 2), count how many times the number 2 appears.

```
In [45]: tup = (1, 2, 2, 3, 2)
         print(tup.count(2))
```

3

i. Finding an Index: In the tuple (10, 20, 30, 40), find the index of the element 30 and print

```
In [46]: tup = (10, 20, 30, 40)
         print(tup.index(30))
```

2